

IE6600 Project 2

ZIP Code Composite Analysis and Interactive Data Dashboard for Prince William County, Virginia

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1. Introduction

The goal of this project is to develop a comprehensive, data-driven dashboard that evaluates the livability, socioeconomic characteristics, and overall desirability of ZIP codes within Prince William County, Virginia. The county presents a unique area of analysis due to both its demographic diversity and its national significance.

A key personal motivation for selecting Prince William County is its connection to the television series *Criminal Minds*. The town of Quantico—located within or adjacent to the county—is home to the FBI Academy and the Behavioral Analysis Unit (BAU), central locations featured throughout the show. As a long-time fan of the series, I was interested in examining the real-world region behind this city and exploring its socioeconomic and safety landscape through data analysis.

2. Data Collection & Sources

At least five variables were required for each ZIP code. This project collected **six** factors from publicly available and authoritative sources:

Variable	Source
Median Household Income	U.S. Census Bureau (ACS 5-Year Estimates)
Population Density	U.S. Census Bureau
Educational Attainment (Bachelor's+)	U.S. Census Bureau
Housing Median Prices	Zillow Research (ZHVI)
Unemployment Rate	Bureau of Labor Statistics
Crime Rate (/1,000 residents)	Prince William County Police Department, ArcGIS Crime Reports

All datasets were aggregated by ZIP code and merged into a final analysis table.

3. Variables Used

Variable	Definition
Median Income	Median household income of the ZIP code
Education Rate	Percentage of residents with a bachelor's degree or higher
Housing Price	Median home value (ZHVI)
Unemployment Rate	Percent of civilians unemployed
Crime Rate	Number of reported crimes per 1,000 residents
Population Density	People per square mile

Each variable was processed numerically and prepared for normalization.

4. Methodology

4.1 Data Cleaning

1. Removed formatting symbols (commas, dollar signs, percentage symbols)
2. Converted all values to numeric format
3. Ensured ZIP codes were 5-digit strings
4. Merged the dataset with geographic boundary files (GeoJSON) for mapping

4.2 Weighting System

Each variable was assigned a weight based on its relevance to regional desirability:

Variable	Weight
Crime Rate (C)	30%
Median Income (I)	20%
Education Level (E)	20%
Housing Prices (H)	10%
Unemployment Rate (U)	10%
Population Density (D)	10%

Explanation:

Crime Rate — Weight: 30%

Crime rate is the most heavily weighted variable because public safety is one of the strongest determinants of livability. Residents generally prioritize safe neighborhoods over any other socioeconomic factor. High crime areas often correlate with lower property values, reduced business activity, and lower perceived quality of life.

Median Income — Weight: 20%

Income is a strong indicator of economic stability and access to resources. ZIP codes with higher median household income generally have better public services, stronger housing markets, and improved quality of life. It also indirectly reflects purchasing power and community investment.

Education Level — Weight: 20%

Educational attainment influences long-term economic mobility, neighborhood development, and overall social outcomes. Areas with higher percentages of bachelor's degree holders tend to have stronger job markets, higher civic participation, and lower crime rates.

It is weighted equally with income due to its comparable significance.

Housing Prices — Weight: 10%

Housing prices reflect market demand and desirability, but they can be influenced by macroeconomic factors outside the local community. Therefore, they are included with a moderate weight to avoid overemphasizing affordability or costliness alone.

Unemployment Rate — Weight: 10%

Employment is crucial for economic sustainability, but unemployment rates can be short-term and fluctuate with broader economic cycles. Thus, it contributes moderately to the final score.

Population Density — Weight: 10%

Density affects congestion, infrastructure strain, and availability of open space. While relevant, it is less critical than income, education, and crime rate—so a smaller weight is applied.

4.3 Composite Score Calculation

FinalScore =

$$30\% \times C + 20\% \times I + 20\% \times E + 10\% \times H + 10\% \times U + \% \times 10D$$

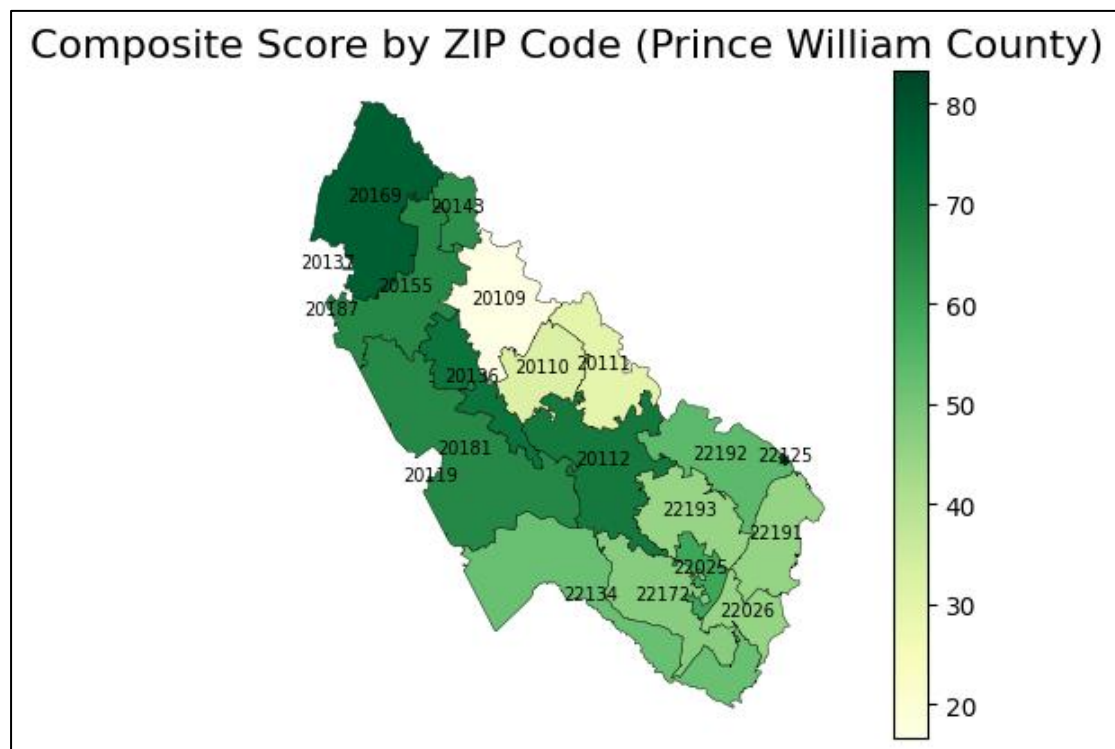
The final score was rescaled to a 0–100 scale for interpretability.

5. Dashboard Description

The interactive dashboard includes the following components:

5.1 Choropleth Map

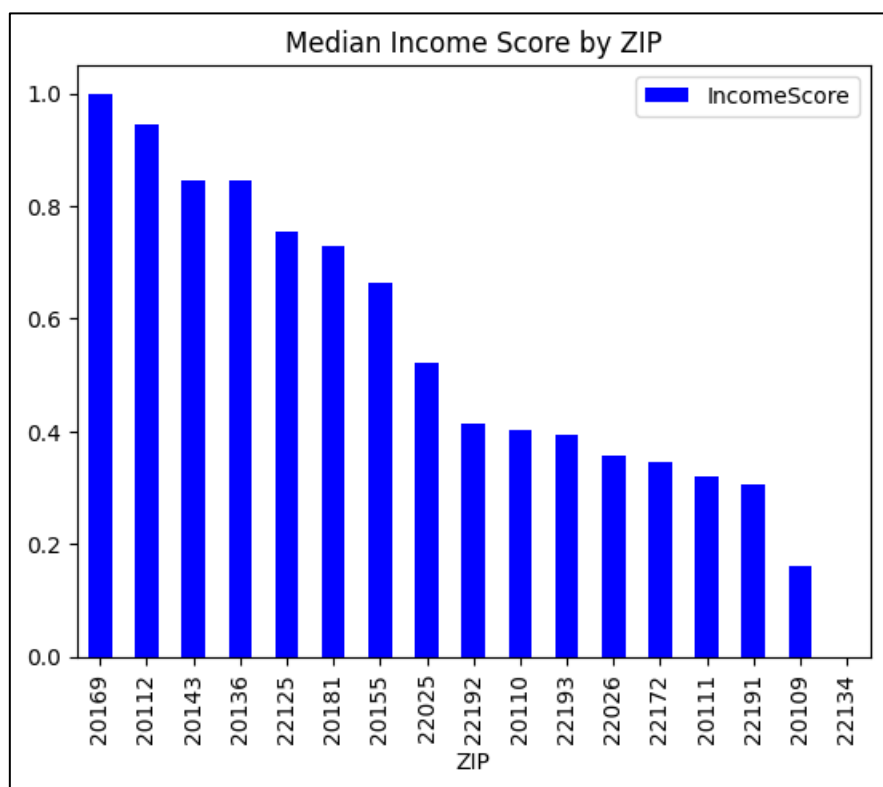
- Visualizes composite score by ZIP code
- Color-coded (yellow → green) to reflect desirability
- Labels embedded on each ZIP polygon



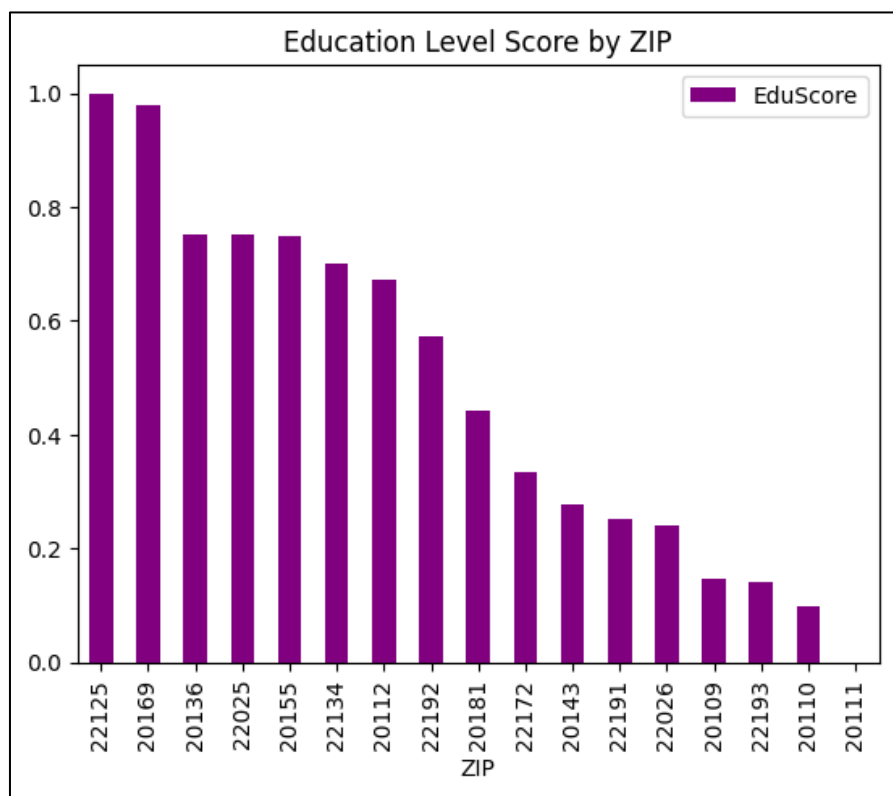
5.2. Variable-Specific Charts

Bar charts for:

- (1) Median income



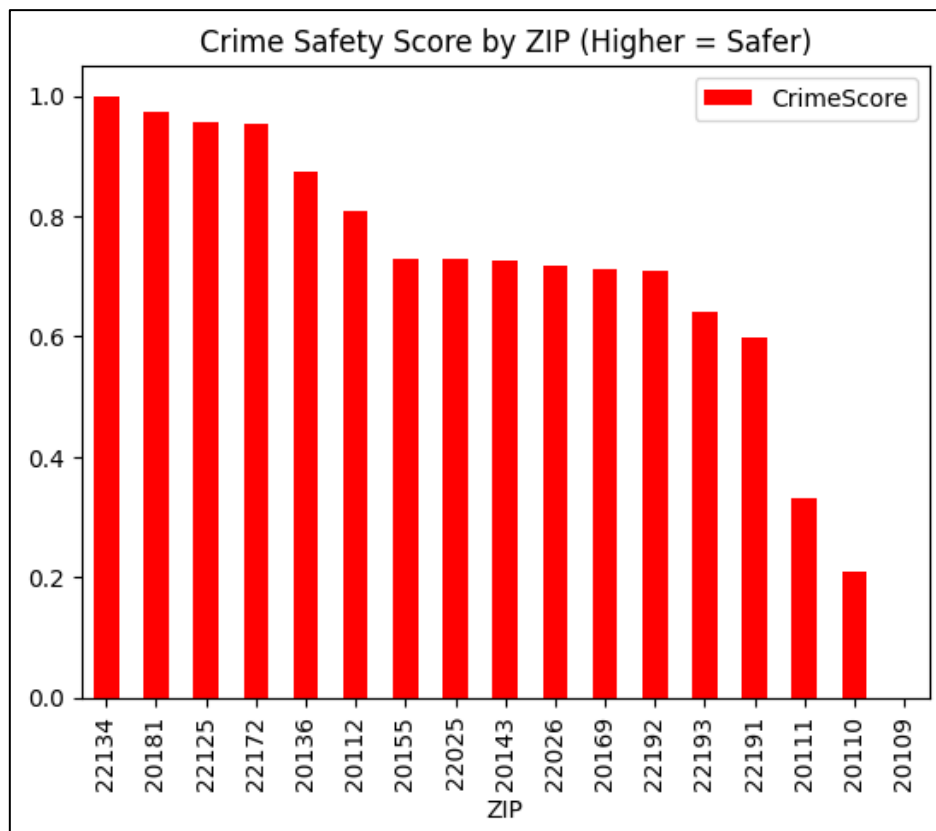
(2) Education level



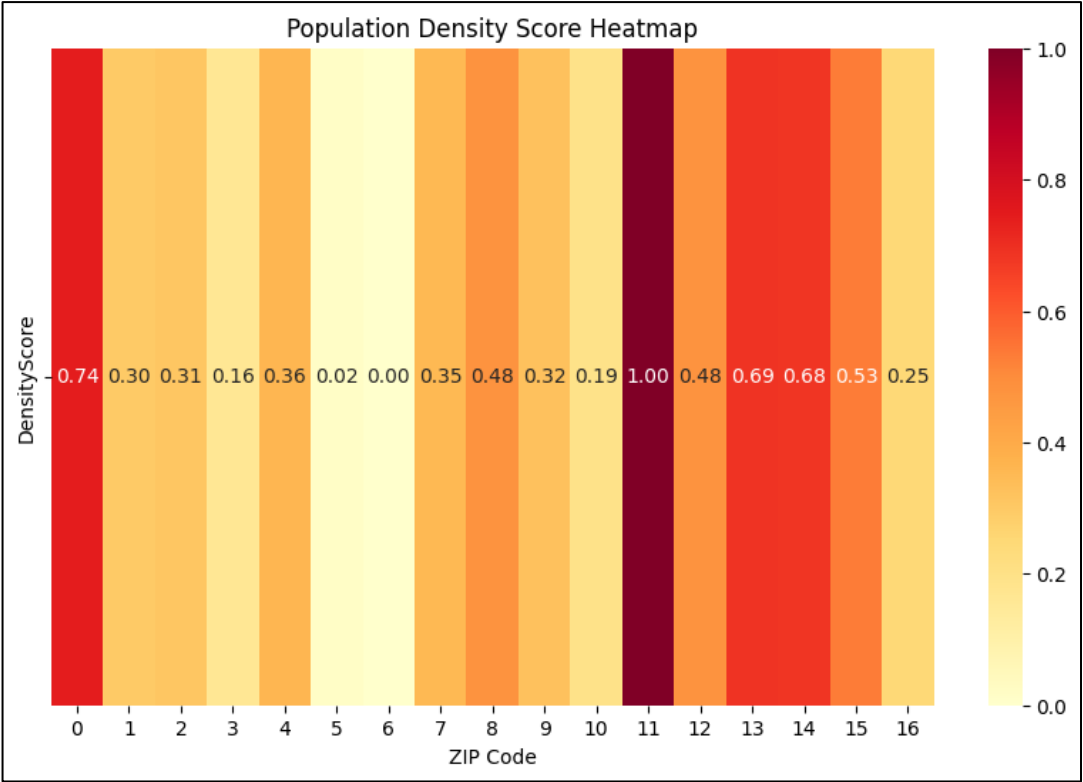
(3) Housing prices



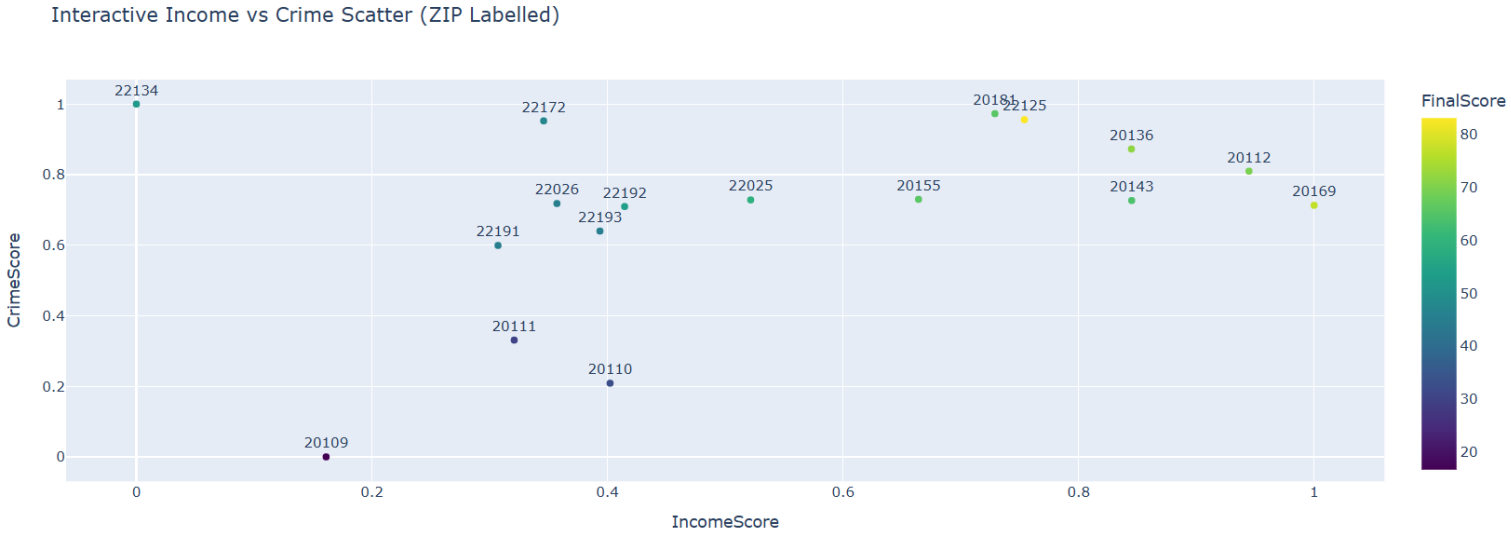
(4) Crime safety score



(5) Heatmap for population density



(6) Scatter plots (Income vs Crime)



5.3 Score Breakdown Interactive Table

- Displays each ZIP code's values across all variables
- Includes ranking and weight references
- Fully interactive (sortable, hover, scrollable)

ZIP Code Score Breakdown

ZIP	Rank	FinalScore	CrimeScore	IncomeScore	EduScore	HousingScore	UnempScore	DensityScore
22125	1	83.21	0.956	0.754	1	0.315	0.887	0.743
20169	2	77.19	0.713	1	0.979	0.559	0.763	0.3
20136	3	71.74	0.873	0.845	0.753	0.443	0.608	0.309
20112	4	69.69	0.81	0.945	0.673	0.482	0.66	0.163
20155	5	65.81	0.73	0.664	0.75	0.5	0.701	0.361
20181	6	65.74	0.973	0.729	0.443	0.553	0.742	0.016
20143	7	64.24	0.727	0.845	0.277	1	1	0
22025	8	59.12	0.728	0.522	0.753	0.331	0.495	0.352
22192	9	53.83	0.71	0.415	0.571	0.29	0.515	0.477
22134	10	52.18	1	0	0.699	0	0.495	0.325
22172	11	47.56	0.952	0.346	0.333	0.347	0	0.194
22191	12	45.38	0.599	0.307	0.253	0.248	0.371	1
22026	13	45.26	0.718	0.357	0.241	0.253	0.443	0.478
22193	14	44.76	0.64	0.394	0.14	0.242	0.557	0.691
20110	15	32.66	0.209	0.402	0.098	0.265	0.691	0.684
20111	16	30.05	0.331	0.321	0	0.27	0.567	0.533
20109	17	16.63	0	0.161	0.146	0.222	0.577	0.25

Weights:
CrimeScore = *30
IncomeScore = *20
EduScore = *20
HousingScore = *10
UnempScore = *10
DensityScore = *10

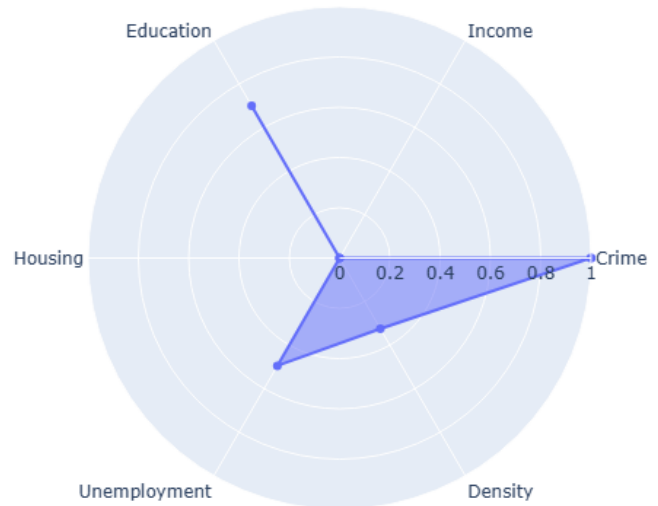
5.4 Radar Chart for Individual ZIP Analysis

- Dropdown selector to compare ZIP characteristics
- Helps visualize strengths and weaknesses of each region

All visualizations were created using Plotly and GeoPandas.

Interactive ZIP Score Breakdown Radar Chart

22134 ▼



6. Results & Insights

6.1 Top-performing ZIP Codes

The highest composite scores were seen in:

1. **22125**
2. **20169**
3. **20136**
4. **20112**

These ZIP codes consistently demonstrate:

- Low crime rates
- High income levels
- Strong educational attainment
- Low unemployment

6.2 Lower-performing ZIP Codes

Lowest scores:

1. **20109**
2. **20111**
3. **20110**

These areas show elevated crime rates, lower educational attainment, and higher density—factors lowering their overall desirability score.

Overall Patterns

- Crime rate and income level correlated strongly with final score.
- ZIP codes near Gainesville and Bristow performed much better than those around Manassas.
- Housing prices alone were not a consistent predictor of desirability.

7. Conclusion

This analysis provides a comprehensive comparison of ZIP codes within Prince William County by integrating socioeconomic, housing, demographic, and public safety indicators into a unified composite scoring model. The resulting dashboard highlights meaningful contrasts across the region and helps reveal factors that shape community desirability.

One particularly interesting finding relates to **Quantico (ZIP 22134)**. Although Quantico is widely recognized as a major federal training hub—home to the FBI Academy and the Behavioral Analysis Unit (BAU) featured in *Criminal Minds*—its composite score is not among the highest in the county. The lower ranking can be attributed to several structural characteristics of the area. Quantico has a relatively small residential population, limited housing availability, and a large portion of land designated for military or federal use, which depresses median income and housing scores. Additionally, its extremely low housing inventory and specialized employment base do not reflect typical suburban community

indicators, resulting in lower performance in some weighted categories despite its national prominence.

On the other hand, the **highest-scoring ZIP codes**—such as **22125 (Triangle)**, **20169 (Haymarket)**, **20136 (Bristow)**, and **20112 (Manassas-area suburbs)**—consistently demonstrate strong socioeconomic profiles. These areas combine high median incomes, strong educational attainment levels, low unemployment, and relatively low crime rates. Many of these ZIP codes represent fast-growing suburban communities with newer housing developments, better school performance metrics, and strong commuter access to Washington, D.C. Their suburban character also contributes to favorable population density scores, which further raises their composite rankings.

In contrast, historically urbanized or commercial-heavy ZIP codes such as **20109**, **20111**, and **20110** rank significantly lower due to higher crime rates, older housing stock, denser populations, and lower education and income indicators. These pockets reflect long-standing regional divides between newer western developments (e.g., Bristow, Haymarket) and older Manassas-area corridors with greater socioeconomic variability.

Overall, this project demonstrates how multi-factor scoring can reveal nuanced differences within a county and identify not only high-performing residential communities but also unique regions like Quantico whose scores reflect structural realities rather than reputation. By combining geographic visualization with an interactive analytical dashboard, the model provides a clear, data-driven perspective for evaluating community desirability across Prince William County.

8. APA References

U.S. Census Bureau. (2023). *American Community Survey 5-Year Estimates*.

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